



Dear OSTP Team & Mr. D’Souza:

Verance Corporation appreciates the opportunity to respond to the Request for Information (RFI) on the Development of the Artificial Intelligence (AI) Action Plan (Federal Register Document No. 2025-02305). Verance is a worldwide leader in watermark technology development, founded in 1995. We have created global standards for durable digital watermark technology for the worldwide recorded music industry, motion picture industry, and broadcast television industries in connection with the DVD-Audio, Blu-ray Disc, and NextGen TV/ATSC 3.0 entertainment formats. Verance’s watermark technology has been included in billions of media assets and consumer products distributed to consumers by leading entertainment and technology companies worldwide.

The AI Action Plan as directed by the Presidential Executive Order of January 23, 2025, has stated goals of defining priority policy actions needed to sustain and enhance America’s AI dominance, and to insure that private sector AI innovation is not hampered by unnecessarily burdensome requirements. To this end, we believe that the following three points should be key considerations in the AI Action Plan:

1. The prevalence of AI-generated disinformation on social media platforms undermines public trust in all media and threatens democratic institutions to the detriment of America’s AI leadership.

2. Provenance authentication systems utilizing digital watermark technology offer a durable, practical means of regaining public trust in media by identifying provenance metadata for both synthetic and non-synthetic media.

3. Private sector efforts at developing technology and standards for provenance authentication provide a path to public trust in an AI dominated media landscape without unduly burdensome government regulations.

Explanation of each of these points follow.

1. The prevalence of AI-generated disinformation on social media platforms undermines public trust and democratic institutions and threatens America's AI leadership.

America's standing as a haven for freedom of expression makes it more vulnerable than most societies to the dangers of misinformation. Today's fractured and chaotic media landscape engendered by social media, multi-varied sources of information, and AI generated misinformation makes it all the more difficult for Americans to know which sources of information to believe and which not to believe.

Deepfakes. The deepfake problem is one example of the harms of AI-generated misinformation. Lately, we've seen an endless flow of stories splashed across newspaper sites and TV screens about the dangers of deepfakes. They range from their use to scare grandparents into sending money to save their relatives to scammers trying to secure jobs at tech companies in order to access sensitive information.

These deepfake threats hit every aspect of society, including national security and our elections – a fact recognized by a bipartisan group of lawmakers. Citing how deepfakes, AI-generated audio and visual content have been used to undermine consumer confidence in everything from commercial products to our nation's media and elections, Congress recently jumped into action.

The lawmakers introduced a bill requiring both the development of standards to identify and label AI-generated content, as well as generative AI developers and online content platforms to provide disclosures on AI-generated content.

This is a great start and arrives at a critical juncture. A recent survey conducted by HarrisX revealed that most U.S. adults were hard-pressed to detect whether AI or an actual person had created certain test video. Open AI also released Sora, a powerful tool that can create realistic videos using a simple text prompt. And the U.S. Government Accountability Office (GAO) issued an update stating that deepfakes — video, audio or images that seem real but were manipulated with AI — have been used to try to interfere in elections.

Lawmakers from both parties, along with the non-partisan GAO, believe that the clear labeling of AI material akin to "nutritional" labeling is necessary to our national security and to maintain an informed

electorate. The public also is calling for AI material to be clearly labeled, with an overwhelming majority of the HarrisX poll affirming the U.S. government must enact those regulations.

To get this right, additional steps are also needed. This includes the GAO's call for systematic changes that would allow all content - human and AI-generated - to be labeled, detected and authenticated in real time.

Broadcast News shared on Social Media. The growing prevalence of AI-generated disinformation on social media platforms is increasingly undermining public trust in media and threatening civil discourse. Consumers place trust in specific sources of information, so a universally accessible and effective method for ensuring that broadcast news is consistently presented to audiences with an assurance of authenticity across all platforms is urgently needed and receiving significant attention from legislative and regulatory bodies. However, a solution that relies solely on labeling of potential harmful AI-generated media by ethical AI platforms will lead the public to assume that unlabeled content from unethical AI-generation platforms is as trustworthy as unlabeled authentic broadcast news. Hence, a solution is needed that addresses non-synthetic as well as synthetic media.

Verance has been in the labeling, i.e. watermarking, industry for over 25 years. Our team has helped develop the watermarking and content labeling technology adopted by standards organizations representing over forty countries worldwide. With that in mind, we see fit to share with Congress and government regulators some valuable lessons learned they should consider when trying to regulate the new frontier of deepfakes and AI-generated content.

First, it is essential to recognize that any solution limited to presenting consumers with provenance labels solely for AI-generated content needs to take into account the most dangerous AI-generated disinformation – that which is produced beyond the reach of U.S. law.

A robust defense against disinformation requires trustworthy content and trustworthy labeling, so the scope of regulatory action should be broad enough to ensure the most important provenance labels – those attached to authentic and official communications such as trusted news outlets and the U.S. government itself – are recognized and presented to the public by social media platforms.

Fortunately, as discussed below, the private sector, through cross-industry collaborations that include technology and media companies in tandem with consumer device makers, have made substantial strides in establishing standards-based solutions for labeling and authenticating all forms of media content.

Mature, globally recognized, consensus-based technology standards for associating tamper-resistant labels with content using watermarking are already in widespread commercial use for media and entertainment including, notably, broadcast news. Additionally, standards that enable tamper-evident metadata for content authentication – information distributed together with content that can be used to determine whether it has been modified – are also advancing rapidly. Public-sector efforts must build on this substantial progress by the private sector.

2. Provenance authentication systems utilizing digital watermark technology offer a durable, practical means of regaining public trust in media by identifying provenance metadata for both synthetic and non-synthetic media.

A fundamental promise of provenance authentication technology is to enable accurate *trust signals* associated with media content to be securely and reliably established by technology platforms and devices for the benefit of users. Trust signals are trustworthy indications of how and by whom the content was created that enable users to make informed decisions about the degree of trust that they will place in the content. Significant emphasis has been placed on ensuring that trust signals associated with synthetic content are available, so that the users can identify this potentially harmful content and its nature into account.

But, of course, no matter how much progress is made towards ensuring that synthetic media incorporates trust signals, it is unavoidable that disinformation agents will continue to have the ability to generate harmful content lacking trust signals, including non-synthetic content using, for example, open source, state sponsored, or other generative technology outside the reach of US policy.

So, what about non-synthetic sources of data that *should* be trusted by the public, such as official government communications and trusted news sources? If this content does not reliably convey trust signals, the public will assume that their absence means the content can be trusted, and such false trust would accrue equally to disinformation.

The strongest defense against harmful synthetic content, is therefore to establish means for all classes of trustworthy content – including synthetic and non-synthetic – to convey authentic, traceable provenance, thereby reducing public trust in content lacking authenticity measures.

This approach parallels that taken on the World Wide Web, where TLS security (the lock icon in the browser bar) provides a widely employed and well-understood trust signal in the identity of the organization operating a website. With supporting public education, its presence can provide consumers with a meaningful trust signal and its absence can be understood to provide a reason for caution.

Digital watermarking. Digital watermarking offers a durable, practical means of identifying provenance metadata. A recent report issued by NIST (“Reducing Risks Posed by Synthetic Content” NIST AI 100-4) provides a substantial discussion of digital watermarking and its use for attaching durable labels to media content¹.

However, the discussion overlooks a highly relevant and widely used application of digital watermarking, which is the use of watermarking to identify metadata *by reference*. In this use of the technology, a digital watermark embedded in a media asset carries information that enables retrieval of metadata associated with the asset from a separate data source; e.g., a database.

Another report, the “Bipartisan House Task force report on Artificial Intelligence” of the 118th Congress recommends that Congress should work with industry to support a standardized ecosystem for technical solutions to synthetic content, such as the standardization of the technical solutions. These technical solutions include the use of watermarking for indicating the origins of content and metadata recording to enable content provenance, and not just synthetic content.²

As described in detail below, the open standards for cryptographically authenticated metadata developed by the Coalition for Provenance and Authenticity (C2PA) and for metadata recovery using audio and video watermarking developed by the Advanced Television Systems Committee (ATSC) together provide a scalable, interoperable means for broadcasters and social media platforms to promote the authenticity of broadcast news content shared on social media.

Trust, broadcast news, and social media. Any attempt to address false information must proceed from an understanding of how people come to place trust in information. The prevalence of information ‘bubbles’ demonstrates that people primarily place trust in specific sources of information. If information appears unaltered and from a trusted source, most people will consider that information to be factual.

Proposed solution to the broadcast news social media use case. The opportunity to verify the authenticity of broadcast news content posted to social media includes success scenarios, where what is ultimately posted is verified and exception scenarios, where what was attempted to be posted is flagged as unverified. To avoid ‘alert fatigue’ it is important to maximize success scenarios whenever practical.

Watermarking addresses the fragility of cryptographic metadata. If uploaded content has been edited or transcoded such that a single bit of the media is altered, the cryptographic hash will change, and the content will not be verifiable. In this case, if the content is also watermarked, the C2PA metadata can be recovered and can be used to retrieve authenticated, canonical content, potentially improving the success scenario of posting validated content. This can be done even if the content was not originally posted by the broadcaster to a website but instead sourced directly from broadcast.

3. Private sector efforts at developing technology and standards for provenance authentication provide a path to public trust in an AI dominated media landscape without unduly burdensome government regulations.

A fundamental element of the provenance challenge is that the global Internet facilitates content following an unconstrained path from its place of creation to that of presentation. This makes interoperability between the systems that generate and validate authenticity signals essential.

Cryptographic metadata standard. On February 22, 2021, Microsoft, the BBC, Adobe, Arm, Intel and Truepic created the Coalition for Content Provenance and Authenticity (C2PA). Today C2PA includes more than 120 companies and continues to grow. Members include Google, Intel, Microsoft, OpenAI, Sony, Truepic, Amazon Web Services, ARM, Canon, Leica, New York Times, Nikon, and NHK.

C2PA provides an open standard to associate tamper-evident cryptographic metadata to any audio-video content, a mechanism to recover that metadata using an embedded watermark in the content and a method for retrieving unaltered content – the ‘canonical content’.

Another standard body, the Content Authenticity Initiative (CAI), advocates for the use of digital watermarking for provenance authentication.

Digital watermark standards. The use of digital watermarking to identify provenance metadata is receiving considerable attention currently. The C2PA incorporates support for the use of digital watermarking to associate provenance manifests with content, and recent research details how that the C2PA provenance authentication and ATSC watermarking standards can be used in combination to attain a fully open and interoperable means for durable provenance authentication of broadcast content.³

A variety of watermark technologies have been proposed to address AI-related content provenance and authentication issues. There is considerable debate among technologists as to how to determine the suitability of any proposed watermark technology to this purpose, including the impact of watermarks on the perceptual quality of media to which it is applied, its resiliency to modifications that occur naturally in the course of media content distribution and modification and its resistance to tampering, removal, forgery, or other security risks.

Verance believes that there is therefore a critical need for the establishment of standard practices in the objective performance evaluation and benchmarking the performance of watermark technologies.

There are a number of existing technical standards specifying the use of watermarks. Some of these standards include performance evaluation and benchmarking frameworks. Verance has assisted a number of Standards Development Organizations in establishing watermark technology performance evaluation and benchmarking frameworks. These include:

- (2015) Society of Motion Picture and Television Engineers (SMPTE) Technology Committee on Television and Broadband (24TB) in connection with their evaluation of watermark technologies submitted in response to the “Request For Proposals for Open Standard to Bind ID to Media Assets” which led to the creation of the *SMPTE 2112-10 Standard: Open Binding of Content Identifiers* standard;
- (2014) ATSC, the US-based international broadcast standards organization, has published a set of standards for this function (ATSC A/334, A/335, and A/336) which have been adopted by numerous television broadcasters and are in widely deployed to enable recovery of timed metadata associated with broadcast video services. The previously mentioned draft NIST AI 100-4 report cites the elements of this standard that specify the physical data transmission layers of this watermark system (A/334 and A/335), but overlooks the highly relevant upper layer specifications (A/336) that enable use of these

watermarks for publication and retrieval of timed metadata from network servers in service of authentication;^{4 5 6} and

- (1999) Secure Digital Music Initiative (SDMI) in connection with their evaluation of watermark technologies submitted in response to the Call for Proposals for Phase I and Phase II Screening Technologies.

In connection with the development of watermarking technical standards for music, film, and television industries, each of these standards organizations have developed objective methodologies and evaluation and assessment of the quality, efficacy, and security of watermark systems offered by responding companies in service of enabling its membership to assess and compare alternative technologies based on objective performance metrics to establish fitness for purpose. In order to establish similar assurances in the context of AI-generated content, we are now faced with a critical need for the establishment of open and accessible performance evaluation criteria for watermarking technologies used in this context.

Verance Corporation Expertise. Verance Corporation is a watermarking technology specialist established in 1995. Verance's watermark technologies have been established as industry standards by the music, film, television, consumer electronics, and computer industries and included in over 500 million consumer products from hundreds of companies worldwide with application to copyright identification and protection, rights management, interactive media experiences, media usage measurement, and addressable advertising.

In connection with the standardization activities, as mentioned above, Verance has served as the principal contributor to multiple industry efforts to establish objective performance benchmarking and evaluation frameworks for watermarking technologies. We believe that any effort to establish guidelines or regulation related to the use of watermarking technology in the context of AI and provenance authentication will benefit from the decades of knowledge and experience that Verance has in assisting industry standards organization and industry consortia in performing this task.

Verance has complete documentation associated with past frameworks developed in industry for performing objective performance evaluation and benchmarking of watermark technology. We are prepared to contribute this historical work, which has already received consensus support among many competing vendors and end-users of watermarking technology, as input to the development of new approaches suited to the current technology environments.

Verance is also a commercial developer of watermark technologies for audio and video and has laboratories, scientists, and commercial watermark technology implementations available for developing and demonstrating watermark technology performance evaluation and benchmarking frameworks. More information can be found at www.verance.com.

Conclusions. Public trust in the media is important to adoption and acceptance of AI technology. The need is acute in the U.S. where government control over the media is minimal. The development of

technology for authentication and provenance determination will be an important element in gaining public trust in a media landscape filled with AI-generated content. This can be done by the private sector without burdensome government regulations in keeping with the goals of the Executive Order. Verance's technology and capabilities can make an important contribution to this effort.

We look forward to continuing engagement with the OSTP and the NSF and thank you for your consideration.

Sincerely,
Nil Shah
Chief Executive Officer

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

¹ NIST AI 100-4 2024. Reducing Risks Posed by Synthetic Content. <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-4.pdf>

² 118th Congress 2024. Bipartisan House Task Force Report on Artificial Intelligence. <https://science.house.gov/2024/12/house-bipartisan-task-force-on-artificial-intelligence-delivers-report>

³ C2PA Specifications v2.0. 2024. Coalition for Content Provenance and Authenticity. <https://c2pa.org/specifications/specifications/2.0/index.html>.

⁴ ATSC A/334:2024. Audio Watermark Emission. Advanced Television Systems Committee. <https://www.atsc.org/atsc-documents/a3342016-audio-watermark-emission/>.

⁵ ATSC A/335:2024. Video Watermark Emission. Advanced Television Systems Committee. <https://www.atsc.org/atsc-documents/a3352016-video-watermark-emission/>.

⁶ ATSC A/336:2024. Content Recovery in Redistribution Scenarios. Advanced Television Systems Committee. <https://www.atsc.org/atsc-documents/a3362017-content-recovery-redistribution-scenarios/>.